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Case Scenario : MySQL Queries Implementation for QR Enabled Automatic Bus Ticket Booking System

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mysql>
mysql> -- EXPERIMENT 25: Case Scenario - QR Enabled Automatic Bus Ticket Booking System
mysql> DROP DATABASE IF EXISTS bus_db;
Query OK, 4 rows affected (0.06 sec)

mysql> CREATE DATABASE bus_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE bus_db;
Database changed
mysql>
mysql> CREATE TABLE Route (
->     route_no INT PRIMARY KEY,
->     source VARCHAR(30) NOT NULL,
->     destination VARCHAR(30) NOT NULL,
->     distance_km INT
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Bus (
->     bus_no INT PRIMARY KEY,
->     bus_name VARCHAR(50) NOT NULL,
->     route_no INT,
->     capacity INT NOT NULL,
->     FOREIGN KEY (route_no) REFERENCES Route(route_no)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Passenger (
->     passenger_id INT PRIMARY KEY,
->     name VARCHAR(50) NOT NULL,
->     phone VARCHAR(15)
-> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> CREATE TABLE Ticket (
  ->   ticket_id INT AUTO_INCREMENT PRIMARY KEY,
  ->   passenger_id INT,
  ->   bus_no INT,
  ->   travel_date DATE NOT NULL,
  ->   qr_code VARCHAR(100) UNIQUE,
  ->   fare DECIMAL(6,2),
  ->   FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id),
  ->   FOREIGN KEY (bus_no) REFERENCES Bus(bus_no)
  -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
mysql> INSERT INTO Route VALUES
  -> (1,'Chennai','Vellore',140),
  -> (2,'Chennai','Trichy',320),
  -> (3,'Chennai','Pondicherry',160);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO Bus VALUES
  -> (101,'TN Express 1',1,50),
  -> (102,'TN Express 2',2,45),
  -> (103,'TN Express 3',3,50);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO Passenger VALUES
  -> (1,'Karthik R','9880011111'),
  -> (2,'Meena S','9880022222'),
  -> (3,'Rahul Dev','9880033333');
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO Ticket (passenger_id, bus_no, travel_date, qr_code, fare) VALUES
  -> (1,101,'2026-08-05','QR-1-101-20260805',150.00),
  -> (2,102,'2026-08-06','QR-2-102-20260806',320.00),
  -> (3,103,'2026-08-05','QR-3-103-20260805',180.00);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> SELECT t.qr_code, p.name, b.bus_name, r.source, r.destination, t.travel_date, t.fare
  -> FROM Ticket t
  -> JOIN Passenger p ON t.passenger_id = p.passenger_id
  -> JOIN Bus b ON t.bus_no = b.bus_no
  -> JOIN Route r ON b.route_no = r.route_no;
```

qr_code	name	bus_name	source	destination	travel_date	fare
QR-1-101-20260805	Karthik R	TN Express 1	Chennai	Vellore	2026-08-05	150.00
QR-2-102-20260806	Meena S	TN Express 2	Chennai	Trichy	2026-08-06	320.00
QR-3-103-20260805	Rahul Dev	TN Express 3	Chennai	Pondicherry	2026-08-05	180.00

3 rows in set (0.00 sec)

```
mysql>
mysql> SELECT b.bus_name, COUNT(*) AS tickets_booked
  -> FROM Ticket t JOIN Bus b ON t.bus_no = b.bus_no
  -> GROUP BY b.bus_name;
```

bus_name	tickets_booked
TN Express 1	1
TN Express 2	1
TN Express 3	1

3 rows in set (0.00 sec)

```
mysql>
mysql> SELECT r.source, r.destination, SUM(t.fare) AS total_revenue
-> FROM Ticket t
-> JOIN Bus b ON t.bus_no = b.bus_no
-> JOIN Route r ON b.route_no = r.route_no
-> GROUP BY r.source, r.destination;
```

source	destination	total_revenue
Chennai	Vellore	150.00
Chennai	Trichy	320.00
Chennai	Pondicherry	180.00

3 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM Ticket WHERE qr_code = 'QR-1-101-20260805';
```

ticket_id	passenger_id	bus_no	travel_date	qr_code	fare
1	1	101	2026-08-05	QR-1-101-20260805	150.00

1 row in set (0.00 sec)